

IMO 2026 — Problem 2

Solution by GPT-5.6 Sol (xhigh reasoning)

2026-07-19

Source: `results/gpt-5.6-sol-xhigh/problem-02/current.md` — autonomous run, no human intervention; solution as written by the model. Multi-round run: earlier-round audit logs in `scratch/`.

Problem

Let ABC be a triangle and let points M and N be the midpoints of sides AB and AC , respectively. Let points K and L be chosen inside triangles BMC and BNC , respectively, such that K lies inside the angle LBA , L lies inside the angle ACK , and $\angle KBA = \angle ACL$, $\angle LBK = \angle LNC$, $\angle LCK = \angle BMK$. Let O be the circumcentre of triangle AKL . Prove that $OM = ON$.

Solution document

Status

solved

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Approaches tried

- **Angle coordinates:** Put $x = \angle KBA = \angle ACL$, $y = \angle LBK = \angle LNC$, and $z = \angle LCK = \angle BMK$. The two small triangles BMK and CNL give explicit coordinates for K and the relation between L and N .
- **Raw Cartesian and complex elimination:** Both verify the theorem, but unorganized expansion produces expressions too long for a rigorous olympiad write-up.
- **Tangent/homogeneous vector factorization:** This gives a compact exact relation between the desired circle determinant and the remaining line-incidence equation.
- **Complex unit-exponential factorization:** This independently gives the same relation. Its one-line “substitute and simplify” proof was too terse.

- **Invariant reorganization (current):** Instead of expanding the coordinates of N , write $L = \lambda N$, reduce both determinants to $K \cdot L$, $[K, L]$, and $|L|^2$, and then compare the four coefficients of the ray parameter. This yields a short invariant expansion and an explicit four-row coefficient check, completing the factorization.
- **Final proof audit (current session):** The argument has been checked through the coordinate setup, ray orientations, the incidence determinant, the power determinant (including its signs), and all exceptional tangent cases. The continuity domain includes all positivity constraints and verifies $\lambda \neq 0$; the coefficient check states an exact reproducible expansion procedure and records every resulting remainder. The factorization lemma is also recorded as a standalone proved lemma, and the Markdown/LaTeX structure has been checked. No mathematical gap or unproved geometric case remains.

Current best

A complete analytic proof is given below. After normalizing $A = 0$, $B = 2$, the three given angle equalities provide explicit coordinates for K, L, N . The remaining incidence condition is $[K - 2N, e^{i(x+z)}N] = 0$. A proved determinant factorization shows that the difference of the powers of M and N with respect to (AKL) is a scalar multiple of this incidence determinant. Hence the powers are equal, which is exactly $OM = ON$.

Full proof

We identify the Euclidean plane with $\mathbb{C}$. After a similarity and, if necessary, a reflection, take

$$A = 0, \quad B = 2, \quad M = 1,$$

with C in the upper half-plane. Lower-case letters will also denote the complex coordinates of the corresponding upper-case points. Put

$$x = \angle KBA = \angle ACL, \quad y = \angle LBK = \angle LNC, \quad z = \angle LCK = \angle BMK.$$

All three numbers are positive. Moreover, the nondegenerate triangles BMK and CNL give

$$0 < x + z < \pi, \quad 0 < x + y < \pi. \quad (1)$$

In particular, the sines occurring below are nonzero.

The conditions $K \in \text{int} \angle LBA$, $\angle KBA = x$, and $\angle LBK = y$ imply that the rays BA, BK, BL occur in this order inside $\angle LBA$. Thus the oriented angle from BA to BL , toward the interior of $\angle ABC$, is $x + y$. Since the vector BA points in direction π , there is a number $t = BL > 0$ such that

$$l = 2 - te^{-i(x+y)}. \quad (2)$$

In triangle BMK , we have $BM = 1$, the angles at B, M are x, z , respectively, and the angle at K is $\pi - x - z$. The sine rule and the direction of MK therefore give

$$k = 1 + \frac{\sin x}{\sin(x+z)} e^{iz}. \quad (3)$$

Also $C = 2N$. In triangle CNL , the angles at C, N are x, y : indeed, C, N, A are collinear and $\angle NCL = \angle ACL = x$, while $\angle LNC = y$. Thus the sine rule gives

$$\frac{NL}{CN} = \frac{\sin x}{\sin(x+y)}.$$

Because L is inside triangle BNC , it lies on the same side of NC as B ; in our orientation this is the clockwise side. Hence the ray NL is obtained from NC by a clockwise turn through y , so

$$l = n + \frac{\sin x}{\sin(x+y)} e^{-iy} n = \lambda n, \quad \lambda = 1 + \frac{\sin x}{\sin(x+y)} e^{-iy}. \quad (4)$$

Finally, because L lies inside $\angle ACK$, the rays CA, CL, CK occur in this order and therefore $\angle ACK = x + z$. The points K and L , being inside triangles BMC and BNC , lie inside ABC ; hence CK is on the interior side of CA . Since $CA = -n$, rotating CA through $x + z$ toward the interior gives the direction $-e^{i(x+z)} n$. Thus $k - 2n$ is a real multiple of $e^{i(x+z)} n$ (the sign of that multiple is immaterial). If

$$[u, v] = \operatorname{Im}(\bar{u}v),$$

this is precisely

$$Q := [k - 2n, e^{i(x+z)} n] = 0. \quad (5)$$

We shall use the following algebraic identity, proved below.

Factorization lemma. Under (2)–(4),

$$\begin{aligned} D &:= [k, l](|n|^2 - 1) - [k, n - 1]|l|^2 + [l, n - 1]|k|^2 \\ &= \frac{\sin x (t \sin x + \sin y)}{\sin(x+y) \sin(x+z)} [k - 2n, e^{i(x+z)} n]. \end{aligned} \quad (6)$$

Assuming the lemma, (5) and (6) yield $D = 0$. It remains to explain why this is the desired conclusion. Let o be the coordinate of the circumcenter O . Since the circle (AKL) passes through the origin, its power function is

$$\mathcal{P}(w) = |w - o|^2 - |o|^2 = |w|^2 - 2 \operatorname{Re}(\bar{o}w).$$

The equations $\mathcal{P}(k) = \mathcal{P}(l) = 0$ are

$$2O \cdot K = |k|^2, \quad 2O \cdot L = |l|^2. \quad (7)$$

The equality $\mathcal{P}(n) = \mathcal{P}(1)$ is equivalently

$$2O \cdot (N - M) = |n|^2 - 1. \quad (8)$$

The augmented determinant of the three linear equations (7)–(8), viewed as equations in the two coordinates of $2O$, is

$$\det \begin{pmatrix} \operatorname{Re} k & \operatorname{Im} k & |k|^2 \\ \operatorname{Re} l & \operatorname{Im} l & |l|^2 \\ \operatorname{Re}(n-1) & \operatorname{Im}(n-1) & |n|^2 - 1 \end{pmatrix}.$$

Expanding it along the third column gives exactly

$$[k, l](|n|^2 - 1) - [k, n - 1]|l|^2 + [l, n - 1]|k|^2 = D.$$

The circumcenter of AKL is given, so A, K, L are noncollinear; equivalently, the first two coefficient vectors in (7) are linearly independent and determine O uniquely. Hence $D = 0$ implies (8). Therefore

$$|ON|^2 - |OM|^2 = \mathcal{P}(n) - \mathcal{P}(1) = 0,$$

and so $OM = ON$.

Verification of the factorization lemma

First suppose that $\cos y \cos z \neq 0$. We have $0 < x < \pi/2$: indeed, the ray BK is inside $\angle ABC$ and the ray CL is inside $\angle ACB$, so x is smaller than both of those angles, while the latter cannot both be at least $\pi/2$. Put

$$p = \tan x, \quad q = \tan y, \quad s = \tan z, \quad r = t \cos x \cos y. \quad (15)$$

The signs of q, s, r will not matter. By (1), $p + q$ and $p + s$ are nonzero. Dividing the coordinate formulas (2)–(4) by the appropriate products of cosines gives

$$K = \left(\frac{2p + s}{p + s}, \frac{ps}{p + s} \right), \quad L = (2 - r(1 - pq), r(p + q)), \quad (16)$$

and

$$L = \lambda N, \quad \lambda = U + iV, \quad U = \frac{2p + q}{p + q}, \quad V = -\frac{pq}{p + q}. \quad (17)$$

Set

$$W = U^2 + V^2, \quad P = K \cdot L, \quad C = [K, L].$$

Since $N = \bar{\lambda}L/W$, we obtain

$$K \cdot N = \frac{UP + VC}{W}, \quad [K, N] = \frac{UC - VP}{W}, \quad |N|^2 = \frac{|L|^2}{W}, \quad [L, N] = -\frac{V|L|^2}{W}. \quad (18)$$

Using $[K, N - 1] = [K, N] + K_y$ and $[L, N - 1] = [L, N] + L_y$ in the definition of D , and then substituting (18), yields

$$WD = C((1 - U)|L|^2 - W) + |L|^2\{V(P - |K|^2) - WK_y\} + WL_y|K|^2. \quad (19)$$

We also replace rotation through $x + z$ by the nonzero real multiple

$$T = (1 - ps) + i(p + s) = \frac{e^{i(x+z)}}{\cos x \cos z}.$$

Put $Q_T = [K - 2N, TN]$. If $T = \alpha + i\beta$, then $[X, TY] = \alpha[X, Y] + \beta X \cdot Y$. Applying this with $\alpha = 1 - ps$, $\beta = p + s$, and using (18), gives

$$WQ_T = C((1 - ps)U + (p + s)V) + P((p + s)U - (1 - ps)V) - 2(p + s)|L|^2. \quad (20)$$

For completeness, we now verify the polynomial identity obtained from (19)–(20). Abbreviate

$$d = p + q, \quad e = p + s, \quad a = 1 - pq, \quad g = (1 + p^2)(1 + q^2), \quad j = \frac{p}{d}.$$

Write $K = (u, v)$. Formula (16) gives the two useful linear relations

$$v = p(2 - u), \quad e(u - 1) = p. \quad (21)$$

Writing $P = P_0 + P_1r$ and $C = C_0 + C_1r$, direct multiplication in (16) gives

$$P_0 = 2u, \quad P_1 = -au + dv, \quad C_0 = -2v, \quad C_1 = du + av, \quad (22)$$

whereas

$$|L|^2 = 4 - 4ar + gr^2, \quad L_y = dr, \quad U = 1 + j, \quad V = -qj, \quad W = 1 + 2j + j^2(1 + q^2). \quad (23)$$

Define

$$X = C_0 + q(P_0 - |K|^2), \quad Y = C_1 + qP_1, \quad Z = -jX - Wv, \quad Z_1 = -jY. \quad (24)$$

From (19) and (22)–(24), collection by powers of r gives

$$WD = A_0 + A_1r + A_2r^2 + A_3r^3, \quad (25)$$

where

$$\begin{aligned} A_0 &= -WC_0 + 4Z, \\ A_1 &= -WC_1 - 4aZ + 4Z_1 + Wd|K|^2, \\ A_2 &= gZ - 4aZ_1, \\ A_3 &= gZ_1. \end{aligned} \quad (26)$$

Similarly, on putting

$$\begin{aligned} \alpha &= (1 - ps)U + eV = (1 - p(e - p))(1 + j) - eqj, \\ \beta &= eU - (1 - ps)V = e(1 + j) + (1 - p(e - p))qj, \end{aligned}$$

formula (20) gives

$$WQ_T = B_0 + B_1r + B_2r^2, \quad (27)$$

where

$$B_0 = C_0\alpha + P_0\beta - 8e, \quad B_1 = C_1\alpha + P_1\beta + 8ea, \quad B_2 = -2eg. \quad (28)$$

Thus

$$deWD = p(q + p(1 + q^2)r)WQ_T \quad (29)$$

is equivalent to the following four coefficient equalities:

$$\begin{aligned} deA_0 &= pqB_0, \\ deA_1 &= p\{qB_1 + p(1 + q^2)B_0\}, \\ deA_2 &= p\{qB_2 + p(1 + q^2)B_1\}, \\ deA_3 &= p^2(1 + q^2)B_2. \end{aligned} \quad (30)$$

Here is an explicit check, included to leave no algebra hidden. Let R_i be the left side minus the right side in the i -th equality in (30), numbered from 0 to 3. Substitution from (21)–(28), using first $v = p(2 - u)$, gives

$$R_0 = \frac{4pq(1 + p^2)}{d}(e(u - 1) - p)(du - 2p - q), \quad (31)$$

$$R_1 = \frac{1 + p^2}{d}(e(u - 1) - p)F_1, \quad R_2 = \frac{p(1 + p^2)(1 + q^2)}{d}(e(u - 1) - p)F_2, \quad R_3 = 0, \quad (32)$$

where

$$\begin{aligned} F_1 &= 5p^3q^2u - 10p^3q^2 + 4p^3u - 8p^3 + 5p^2q^3u - 4p^2q^3 + 4p^2qu \\ &\quad + pq^2u + 2pq^2 + q^3u, \\ F_2 &= p^3qu - 2p^3q + p^2q^2u + 4p^2 + pqu + 2pq + q^2u. \end{aligned}$$

To verify (31)–(32) without any concealed division, multiply each R_i first by d and insert

$$v = p(2 - u), \quad U = 1 + \frac{p}{d}, \quad V = -\frac{pq}{d}, \quad W = 1 + \frac{2p}{d} + \frac{p^2(1 + q^2)}{d^2},$$

together with (22)–(28); collecting the coefficients of u gives exactly the displayed products. This is ordinary distributive expansion, and the formulas (31)–(32) record the full remainders after collection. The second relation in (21) makes every one of them zero, proving (30), hence (29).

Because $W \neq 0$ and $de \neq 0$, (29) gives

$$D = \frac{p\{q + p(1 + q^2)r\}}{de} Q_T. \quad (33)$$

Now

$$Q_T = \frac{Q}{\cos x \cos z}$$

and, using (15),

$$\frac{p\{q + p(1 + q^2)r\}}{(p + q)(p + s) \cos x \cos z} = \frac{\sin x (t \sin x + \sin y)}{\sin(x + y) \sin(x + z)}. \quad (34)$$

Indeed, substitute $p = \sin x / \cos x$, $q = \sin y / \cos y$, $s = \sin z / \cos z$, and $r = t \cos x \cos y$; cancellation gives (34). Equations (33)–(34) prove (6) when $\cos y \cos z \neq 0$.

Finally, both sides of (6), with k, l, n, λ defined by (2)–(4), are continuous functions of (x, y, z, t) on the open set

$$0 < x < \frac{\pi}{2}, \quad y > 0, \quad z > 0, \quad x + y < \pi, \quad x + z < \pi, \quad t > 0.$$

(Here $0 < y < \pi - x$, so $\operatorname{Im} \lambda = -\frac{\sin x \sin y}{\sin(x + y)} < 0$; in particular $\lambda \neq 0$.) The subset on which $\cos y \cos z \neq 0$ is dense in this open set. Thus (6) also holds when $\cos y \cos z = 0$, by continuity. This completes the proof of the factorization lemma and of the theorem.